
IT205: Data Structures

Academic Year 2026-26 (Winter)

Instructor:	Dr. Arpit Rana / Dr. Supantha Pandit [fname_lname@dau.ac.in]
Prerequisites:	Discrete Mathematics, Introduction to Programming (C)
Slot/ Category:	B.Tech. II Semester - Section A / B Core
Course Credits:	3 Credits (3-0-0)
Associated Lab Course:	IT206 / Data Structures Lab using OOP / 2 (1-0-2).
Foundation for:	CS / IT stream of courses
Lectures (1x3) LT-02:	TBD
TA contact info:	TBD

Course Description:

This course aims to introduce the concept of data structures, their indispensability in implementing algorithms, and how they enhance performance. Extensive coverage of the well-known and essential data structures, such as arrays, linked lists, queues, stacks, hash tables, binary search trees, AVL trees, red-black trees, 2-3-4 trees, and heaps, along with associated routines/algorithms such as dictionary operations, order statistics, sorting, will constitute the contents of the course.

Suggested Books:

- 'Data Structures and Algorithms,' by Aho, Hopcroft, and Ullman, Addison-Wesley, 1999. [Textbook]
- 'Data Structures and Algorithms in C++/Java,' by Goodrich, Tamassia, and Goldberg, Wiley, 2011.
- 'Introduction to Algorithms,' by Cormen, Leiserson, Rivest, and Stein, PHI, 2010

Course Outcomes:

After successful completion of the course, the student will have the ability to -

- Utilize appropriate simple data structures in problem-solving
- Understand data abstraction and order statistics
- Understand program efficiency through analysis of algorithms

P1	P2	P3	P4	P5	P6	P7	P8	P9	P10	P11	P12
X	X	X		X				X			X

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Evaluation Scheme (Tentative):

- In-Semester Exams (I & II): 40% (20% + 20%)
- Decipher-o-Cipher (3): 20%
- End-Semester Exam: 40%

Tentative Course Plan¹:

Units	Topics	Number of Lectures
Preliminaries	Representation of data on a computer, data types & array and linked list representations, ways of representing programs and associated data on computers	3
Analysis Tools	The notion of the running time of an algorithm, Recurrences, and Parameters of Performance	3
Dictionary Operations	Find, Max, Min, Successor, Predecessor (query operations); Insert, Delete (modify operations)	6
List Data	Stacks, queues, and variants implementation using arrays and linked lists, as well as hashing.	6
Sorting	Comparison-based sorting algorithms, Other sorting algorithms, lower bounds for comparison-based sorting algorithms, Best-case, worst-case, and average-case running times. Heap Sort, Bubble Sort, etc.	9
Order Statistics	Maximum and minimum elements of a set, finding median, searching for an element of a given rank, finding the rank of a given element, ranks of a subset of elements, maintaining rank information for a dynamic set	3
Trees	Heaps, Binary search trees (BST), the height of BST, Tries, Balanced BSTs: Red Black trees, AVL Trees, B Trees.	9
Graphs	Representation using adjacency matrices and adjacency lists, Graph searching algorithms, BFS, and DFS	3

1 — This course plan is subject to change without notice.

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Preliminary Schedule of the Course:

Week	Lecture	Capstone ¹
Week-1 [5 Jan 2026]	Course Admin: Introduction to Data Structures: Variables, Data Types, and Data Structures.	–
Week-2 [12 Jan 2026]	Algorithm Analysis: Counting operations; Asymptotic Analysis Big-O notation (Upper-Bound); three cases of analysis: Best-case, Worst-case, Average-case; Abstract Data Types (ADTs)	–
Week-3 [19 Jan 2026]	Arrays: Definition, Types, Terminology, Computer Representation, Operations, and Applications.	–
Week-4 [26 Jan 2026]	Linked List: Definition, Types, Terminology, Computer Representation, Operations, and Applications.	–
Week-5 [2 Feb 2026]	Order Statistics	–
Week-6 [9 Feb 2026]	First In-Semester Examination (9 th Feb - Friday to 12 th Feb - Tuesday)	–
Week-6 [16 Feb 2026]	Stacks: Definition, Types, Terminology, Computer Representation, Operations, and Applications.	Decipher-o -Cipher I
Week-7 [23 Feb 2026]	Queues: Definition, Types, Terminology, Computer Representation, Operations, and Applications.	–
Week-8 [2 Mar 2026]	In-semester Break (Entire week)	–
Week-9 [9 Mar 2026]	Tables: Definition, Types, Terminology, Computer Representation, Operations, and Applications.	–
Week-10 [16 Mar 2026]	Second In-Semester Examination (18 th Mar Wednesday to 23 rd Mar Monday)	–

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Week-11 [23 Mar 2026]	Introduction to Trees, Binary Trees, and their Types, Implementation of Binary Trees in Arrays, and Traversals: Inorder, preorder, and postorder; Basic Operations in Binary Trees	Decipher-o-Cipher
Week-12 [30 Mar 2026]	Introduction to BST, BST traversals and sorting, Insertion and creation of BSTs; Deletion in BST, augmented BST, update cost in augmented BST	–
Week-13 [6 Apr 2026]	Introduction to AVL trees: Min and max height of AVL trees, searching on an AVL tree, insertion in an AVL tree	–
Week-14 [13 Apr 2026]	Introduction to heaps, insertion and deletion in heaps, heapsort, heapify, priority queues,	-
Week-15 [20 Apr 2026]	Introduction to Graphs: Representation using adjacency matrices and adjacency lists, Graph searching algorithms: BFS and DFS.	Decipher-o-Cipher
Week-16 [27 Apr 2024]	Last date of Classes, Labs, and Tutorials (29 th Apr 2026)	-
End-Semester Examination (30 th Apr 2026 to 8 th May 2026)		

1 – Decipher-o-Cipher exercise will be assigned group-wise (2 students in each group). The best two scores will be considered.